

上半：解方程

1 线性方程组 $Ax=b$

(一) $n \times n$: 直接法

▷ 三角形方程组

下三角

$$Lx = b$$

$$x_i = \frac{1}{l_{ii}} \left(b_i - \sum_{j=1}^{i-1} l_{ij} x_j \right), \quad i = 1, \dots, n.$$

上三角

$$Ux = b$$

$$x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j=i+1}^n u_{ij} x_j \right), \quad i = n, \dots, 1.$$

复杂度: $O(n^2)$.

▷ Gauss 消元法

$$l_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}, \quad i > k.$$

$$a_{kk}^{(k)} \neq 0, \quad a_{kk}^{(k)} \rightarrow 0 \Rightarrow \text{选主元}.$$

▷ 平凡的 LU

规定

$$\left\{ \begin{array}{l} L = \begin{bmatrix} 1 & & \\ * & 1 & \\ * & * & \ddots \end{bmatrix}, \\ U = \begin{bmatrix} * & * & * \\ & * & * \\ & & \ddots \end{bmatrix}. \end{array} \right.$$

规定 L 为单位下三角, 则

$$A = LU \iff \det A_{1:k,1:k} \neq 0, \quad k = 1, \dots, n.$$

L 单位下三角 $\Rightarrow$ 唯一,

$A = LU = (LD)(D^{-1}U) \Rightarrow$ 不加约束时不唯一.

▷ 一步消元与 Schur 补

$$A = \begin{bmatrix} \alpha & r^\top \\ c & B \end{bmatrix}, \quad \alpha \neq 0$$

$$A = \begin{bmatrix} 1 & 0 \\ c/\alpha & I \end{bmatrix} \begin{bmatrix} \alpha & r^\top \\ 0 & B - \frac{cr^\top}{\alpha} \end{bmatrix} \cdot S = B - \frac{cr^\top}{\alpha}$$

一步消元 $\iff$ Schur 补 S .

▷ 列主元 LU

无主元消元可能遇到

$$a_{kk}^{(k)} = 0 \quad \text{或} \quad |a_{kk}^{(k)}| \ll 1.$$

第 k 步选

$$p = \arg \max_{i \geq k} |a_{ik}^{(k)}|,$$

交换第 k, p 行:

$$PA = LU.$$

$$\text{可逆 } A \Rightarrow PA = LU, \quad |l_{ik}| \leq 1.$$

▷ 完全主元 LU

第 k 步在剩余子阵中选

$$(p, q) = \arg \max_{i, j \geq k} |a_{ij}^{(k)}|,$$

交换第 k, p 行与第 k, q 列:

$$PAQ = LU.$$

行交换改方程顺序, 列交换改未知量顺序.

$$LUz = Pb, \quad x = Qz.$$

▷ Cholesky 分解

$$A = A^\top > 0 \iff \det A_{1:k,1:k} > 0, \quad k = 1, \dots, n$$

$$\iff A = LL^\top, \quad l_{ii} > 0.$$

$$A = A^\top > 0 \Rightarrow \text{主元} > 0, \text{ 无需选主元}.$$

▷ 逐列计算公式

$$A = LL^\top, \quad a_{ij} = \sum_{k=1}^{\min(i,j)} l_{ik} l_{jk}.$$

$$l_{jj} = \sqrt{a_{jj} - \sum_{k=1}^{j-1} l_{jk}^2}, \quad l_{ij} = \frac{a_{ij} - \sum_{k=1}^{j-1} l_{ik} l_{jk}}{l_{jj}}, \quad i > j.$$

根号内 = 第 j 个主元, $\leq 0 \Rightarrow A$ 非正定或误差过大.

$$\text{复杂度: } \frac{1}{3}n^3 + O(n^2).$$

$$Ax = b: \quad Ly = b, \quad L^\top x = y.$$

▷ **LDL^T 分解**

$$A = LDL^T,$$

$$L \text{ 单位下三角}, \quad D = \text{diag}(d_1, \dots, d_n).$$

$$d_j = a_{jj} - \sum_{k=1}^{j-1} l_{jk}^2 d_k, \quad l_{ij} = \frac{a_{ij} - \sum_{k=1}^{j-1} l_{ik} d_k l_{jk}}{d_j}, \quad i > j.$$

$$A > 0 \Rightarrow d_i > 0, \quad A = (LD^{1/2})(LD^{1/2})^T.$$

$$P^T A P = LDL^T, \quad D: 2 \times 2 \text{ 块也可}.$$

▷ **三对角矩阵的追赶法**

三对角追赶法 = 三对角专用 Gauss.

$$\tilde{b}_i \neq 0.$$

$$\begin{bmatrix} b_1 & c_1 & & \\ a_2 & b_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_n & b_n \end{bmatrix} x = d,$$

$$\begin{cases} \tilde{b}_1 = b_1, & \tilde{d}_1 = d_1, \\ \ell_i = \frac{a_i}{\tilde{b}_{i-1}}, \\ \tilde{b}_i = b_i - \ell_i c_{i-1}, \\ \tilde{d}_i = d_i - \ell_i \tilde{d}_{i-1}, \quad i = 2, \dots, n. \end{cases}$$

$$x_n = \frac{\tilde{d}_n}{\tilde{b}_n}, \quad x_i = \frac{\tilde{d}_i - c_i x_{i+1}}{\tilde{b}_i}, \quad i = n-1, \dots, 1.$$

复杂度: $O(n)$.

$$\tilde{b}_i \neq 0.$$

(二) 范数▷ **定义**

$$\begin{cases} \|x\|_1 = \sum_i |x_i|, \\ \|x\|_2 = \sqrt{\sum_i |x_i|^2}, \\ \|x\|_\infty = \max_i |x_i|. \end{cases}$$

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|.$$

$$\begin{cases} \|A\|_1 = \max_j \sum_i |a_{ij}|, \\ \|A\|_\infty = \max_i \sum_j |a_{ij}|, \\ \|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^T A)}, \\ \|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2} = \sqrt{\text{tr}(A^T A)}. \end{cases}$$

▷ **比较**

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1.$$

$$\|x\|_1 \leq \sqrt{n} \|x\|_2, \quad \|x\|_2 \leq \sqrt{n} \|x\|_\infty.$$

$$1 \leq p \leq q \leq \infty \implies \|x\|_q \leq \|x\|_p \leq n^{1/p-1/q} \|x\|_q.$$

$$\forall \|\cdot\|_a, \|\cdot\|_b, \quad \exists c, C > 0: \quad c \|x\|_a \leq \|x\|_b \leq C \|x\|_a.$$

$$\|Ax\| \leq \|A\| \|x\|, \quad \|AB\| \leq \|A\| \|B\|.$$

$$\|A\|_2 \leq \|A\|_F \leq \sqrt{\text{rank}(A)} \|A\|_2, \quad \|AB\|_F \leq \|A\|_F \|B\|_F.$$

▷ **等高线**

$$\|x\|_2 = \text{圆}, \quad \|x\|_1 = \text{菱形}, \quad \|x\|_\infty = \text{方形}.$$

$$\|A\| = \max_{\|x\|=1} \|Ax\|.$$

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}, \quad A = A^T > 0 \implies \kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}.$$

▷ **误差分析里的范数**

$$\rho(A) = \max_i |\lambda_i(A)|, \quad \rho(A) \leq \|A\|.$$

$$A^T A = A A^T \implies \|A\|_2 = \max_i |\lambda_i(A)|.$$

$$\min_{A+\Delta A \text{ 奇异}} \frac{\|\Delta A\|_2}{\|A\|_2} = \frac{1}{\kappa_2(A)}.$$

上式要求 A 可逆.

$$\text{一阶: } \frac{\|\Delta x\|}{\|x\|} \lesssim \kappa(A) \left(\frac{\|\Delta A\|}{\|A\|} + \frac{\|\Delta b\|}{\|b\|} \right).$$

▷ **稳定性与误差分析**

复数情形: $^T \rightarrow *$.

▷ **浮点运算模型**

$$\text{fl}(a \circ b) = (a \circ b)(1 + \delta), \quad |\delta| \leq u.$$

$$\circ \in \{+, -, \times, /\}.$$

$$\prod_{i=1}^k (1 + \delta_i)^{\rho_i} = 1 + \theta_k, \quad |\delta_i| \leq u, \quad |\rho_i| \leq 1,$$

$$|\theta_k| \leq \gamma_k, \quad \gamma_k = \frac{ku}{1 - ku}.$$

$$\gamma_k \approx ku.$$

▷ 矩阵运算误差

$$\mathfrak{fl}(\alpha A) = \alpha A + E, \quad |E| \leq u |\alpha A|,$$

$$\mathfrak{fl}(A + B) = A + B + E, \quad |E| \leq u |A + B|,$$

$$\mathfrak{fl}(AB) = AB + E, \quad |E| \leq \gamma_n |A| |B|.$$

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p}.$$

$$\mathfrak{fl}(x^\top y) = x^\top y + \varepsilon, \quad |\varepsilon| \leq \gamma_n |x|^\top |y|.$$

误差 $\sim u + \text{步数} + \text{消去}$.

▷ 前向误差与后向误差

$$\frac{\|\hat{x} - x\|}{\|x\|},$$

$$(A + \Delta A)\hat{x} = b + \Delta b$$

$$A(\hat{x} - x) = \Delta b - \Delta A \hat{x} \implies \frac{\|\hat{x} - x\|}{\|x\|} \leq \kappa(A) \left(\frac{\|\Delta b\|}{\|b\|} + \frac{\|\Delta A\| \|\hat{x}\|}{\|A\| \|x\|} \right).$$

$$\kappa(A) = \|A\| \|A^{-1}\|.$$

$$b \neq 0, \quad x \neq 0.$$

$$\delta_A = \frac{\|\Delta A\|}{\|A\|}, \quad \delta_b = \frac{\|\Delta b\|}{\|b\|},$$

$$\kappa(A) \delta_A < 1,$$

$$\frac{\|\hat{x} - x\|}{\|x\|} \leq \frac{\kappa(A)}{1 - \kappa(A) \delta_A} (\delta_A + \delta_b).$$

▷ 列主元高斯消元的稳定性

$$PA \approx \hat{L}\hat{U},$$

$$PA + \Delta A = \hat{L}\hat{U}, \quad |\Delta A| \leq \gamma_n |\hat{L}| |\hat{U}|.$$

$$(T + \Delta T)\hat{x} = b, \quad |\Delta T| \leq \gamma_n |T|.$$

$$(A + \Delta A)\hat{x} = b.$$

▷ 增长因子

$$|l_{ij}| \leq 1,$$

$$\rho = \frac{\max_{i,j} |u_{ij}|}{\max_{i,j} |a_{ij}|}.$$

$$\frac{\|\Delta A\|_\infty}{\|A\|_\infty} \lesssim n^3 \rho u.$$

列主元管 L , ρ 管 U .

▷ 逆矩阵扰动

$$\tilde{A} = A + \Delta A.$$

$$\|A^{-1} \Delta A\| < 1 \implies \tilde{A} = A(I + A^{-1} \Delta A) \text{ 可逆},$$

$$\tilde{A}^{-1} = (I + A^{-1} \Delta A)^{-1} A^{-1}.$$

$$(I + E)^{-1} = I - E + E^2 - E^3 + \cdots, \quad \|E\| < 1,$$

$$\implies \|(I + A^{-1} \Delta A)^{-1}\| \leq \frac{1}{1 - \|A^{-1} \Delta A\|},$$

$$\|\tilde{A}^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1} \Delta A\|}.$$

▷ 线性方程组扰动

$$Ax = b, \quad (A + \Delta A)(x + \Delta x) = b + \Delta b.$$

$$\delta_A = \frac{\|\Delta A\|}{\|A\|}, \quad \delta_b = \frac{\|\Delta b\|}{\|b\|},$$

$$\kappa(A) = \|A\| \|A^{-1}\|.$$

$$\kappa(A) \delta_A < 1,$$

$$\frac{\|\Delta x\|}{\|x\|} \leq \frac{\kappa(A)}{1 - \kappa(A) \delta_A} (\delta_A + \delta_b).$$

$$\frac{\|\Delta x\|}{\|x\|} \lesssim \kappa(A) (\delta_A + \delta_b).$$

▷ 残差与真实误差

$$r = b - A\hat{x}, \quad e = x - \hat{x}.$$

$$Ae = r, \quad e = A^{-1}r \implies \frac{\|e\|}{\|x\|} \leq \kappa(A) \frac{\|r\|}{\|b\|}.$$

(三) $m \times n$: 最小二乘问题

▷ 最小二乘问题

$$x_* = \arg \min_x \|Ax - b\|_2, \quad r = b - Ax_*, \quad A^\top r = 0.$$

$$A^\top r = 0 \implies A^\top Ax_* = A^\top b.$$

$$A \text{ 满列秩} \implies x_* = (A^\top A)^{-1} A^\top b \text{ 唯一}.$$

$$\kappa_2(A^\top A) = \kappa_2(A)^2.$$

▷ Moore–Penrose 广义逆

$$A = U\Sigma V^\top, \quad A^+ = V\Sigma^+ U^\top,$$

$$\sigma_i \neq 0 \implies \sigma_i^+ = \frac{1}{\sigma_i}.$$

$$x_* = A^+ b.$$

$$A^+ = (A^\top A)^{-1} A^\top \quad (\text{满列秩}),$$

$$A^+ = A^\top (AA^\top)^{-1} \quad (\text{满行秩}).$$

▷ **Householder 变换**

$$Hx = \alpha e_1.$$

$$\alpha = -\text{sign}(x_1)\|x\|_2, \quad v = x - \alpha e_1,$$

$$x \neq 0, \quad \text{sign}(0) = 1.$$

$$H = I - 2 \frac{vv^\top}{v^\top v}.$$

$$H^\top = H, \quad H^{-1} = H, \quad H^\top H = I.$$

▷ **Givens 旋转**

$$\begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} x_i \\ x_k \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix},$$

$$r = \sqrt{x_i^2 + x_k^2}, \quad c = \frac{x_i}{r}, \quad s = \frac{x_k}{r}.$$

$$r \neq 0.$$

r 常用 hypot.

▷ **QR 求最小二乘**

$$m \geq n, \quad \text{rank}(A) = n.$$

$$A = Q \begin{bmatrix} R \\ 0 \end{bmatrix}, \quad Q^\top Q = I, \quad Q^\top b = \begin{bmatrix} c \\ d \end{bmatrix}.$$

$$\|Ax - b\|_2^2 = \left\| \begin{bmatrix} R \\ 0 \end{bmatrix} x - \begin{bmatrix} c \\ d \end{bmatrix} \right\|_2^2 = \|Rx - c\|_2^2 + \|d\|_2^2.$$

$$\|Ax - b\|_2^2 = \|Rx - c\|_2^2 + \|d\|_2^2$$

$$\implies Rx_* = c, \quad \min_x \|Ax - b\|_2 = \|d\|_2.$$

(四) 迭代法

$$x^{(k+1)} = Bx^{(k)} + f.$$

$$\forall x^{(0)} \text{ 收敛} \iff \rho(B) < 1.$$

$$\|B\| < 1 \Rightarrow \rho(B) < 1.$$

▷ **Jacobi 迭代**

$$A = D - L - U, \quad D: \text{对角}, -L, -U: \text{严格下、上三角}.$$

$$Dx^{(k+1)} = (L + U)x^{(k)} + b,$$

$$x^{(k+1)} = D^{-1}(L + U)x^{(k)} + D^{-1}b.$$

$$B_J = D^{-1}(L + U), \quad \rho(B_J) < 1.$$

▷ **Gauss-Seidel 迭代**

$$(D - L)x^{(k+1)} = Ux^{(k)} + b,$$

$$x^{(k+1)} = (D - L)^{-1}Ux^{(k)} + (D - L)^{-1}b.$$

$$B_{GS} = (D - L)^{-1}U, \quad \rho(B_{GS}) < 1.$$

▷ **SOR 迭代**

$$(D - \omega L)x^{(k+1)} = [(1 - \omega)D + \omega U]x^{(k)} + \omega b.$$

$$\omega = 1 \Rightarrow \text{GS}, \quad 0 < \omega < 1: \text{欠松弛}, \quad \omega > 1: \text{超松弛}.$$

$$B_\omega = (D - \omega L)^{-1}[(1 - \omega)D + \omega U].$$

$$A = A^\top > 0, \quad 0 < \omega < 2 \Rightarrow \rho(B_\omega) < 1.$$

(五) 共轭梯度法

$$A = A^\top > 0, \quad Ax = b.$$

$$\phi(x) = \frac{1}{2}x^\top Ax - b^\top x.$$

$$\nabla \phi(x) = Ax - b,$$

$$\nabla \phi(x) = 0 \iff Ax = b.$$

$$x_* = A^{-1}b, \quad e = x - x_*.$$

$$\phi(x) - \phi(x_*) = \frac{1}{2}e^\top Ae = \frac{1}{2}\|e\|_A^2,$$

$$\|e\|_A = \sqrt{e^\top Ae}.$$

$$Ax = b \iff \min_x \phi(x).$$

▷ **一维精确搜索**

$$x_{k+1} = x_k + \alpha_k p_k.$$

$$r_k = b - Ax_k = -\nabla \phi(x_k).$$

$$\frac{d}{d\alpha} \phi(x_k + \alpha p_k) = 0 \implies \alpha_k = \frac{p_k^\top r_k}{p_k^\top A p_k}.$$

$$p_k^\top r_{k+1} = 0.$$

▷ **最速下降法**

$$p_k = r_k,$$

$$\alpha_k = \frac{r_k^\top r_k}{r_k^\top A r_k},$$

$$x_{k+1} = x_k + \alpha_k r_k,$$

$$r_{k+1} = r_k - \alpha_k A r_k.$$

$$r_0 = b - Ax_0,$$

$$\alpha_k = \frac{r_k^\top r_k}{r_k^\top A r_k},$$

$$x_{k+1} = x_k + \alpha_k r_k,$$

$$r_{k+1} = r_k - \alpha_k A r_k.$$

$$r_{k+1}^\top r_k = 0.$$

通常只相邻正交.

$$\|e_{k+1}\|_A \leq \frac{\kappa_2(A) - 1}{\kappa_2(A) + 1} \|e_k\|_A.$$

▷ 共轭方向

$$p_i^\top A p_j = 0, \quad i \neq j,$$

A -共轭.

$$(A^{1/2} p_i)^\top (A^{1/2} p_j) = 0,$$

▷ 共轭梯度法

$$r_0 = b - A x_0, \quad p_0 = r_0$$

$$k = 0, 1, \dots$$

$$\alpha_k = \frac{r_k^\top r_k}{p_k^\top A p_k},$$

$$x_{k+1} = x_k + \alpha_k p_k,$$

$$r_{k+1} = r_k - \alpha_k A p_k,$$

$$\beta_k = \frac{r_{k+1}^\top r_{k+1}}{r_k^\top r_k},$$

$$p_{k+1} = r_{k+1} + \beta_k p_k.$$

存储: $x_k, r_k, p_k, A p_k$, 每步主耗时: 1 次 matvec.

▷ 正交结构

$$r_i^\top r_j = 0, \quad i \neq j,$$

$$p_i^\top A p_j = 0, \quad i \neq j,$$

$$r_k^\top p_j = 0, \quad j < k.$$

$$p_k^\top r_k = r_k^\top r_k \neq 0.$$

不能写 $p \perp r$.

▷ 有限终止与收敛速度

CG 至多 n 步得到精确解.

$$\text{步数} \leq m_{A, r_0};$$

m_{A, r_0} : 关于 r_0 的最小多项式次数.

A 仅有 s 个不同特征值 $\implies$ 至多 s 步.

CG 的经典误差界为

$$\|e_k\|_A \leq 2 \left(\frac{\sqrt{\kappa_2(A)} - 1}{\sqrt{\kappa_2(A)} + 1} \right)^k \|e_0\|_A.$$

▷ 适用范围

$$A = A^\top > 0.$$

$$A = A^\top > 0 \implies p_k^\top A p_k > 0.$$

α_k 存在, ϕ 严格凸.

对称不定: MINRES, 非对称: GMRES, BiCGStab.

▷ 预条件

$$M = M^\top > 0, \quad M \approx A,$$

$$\kappa_2(M^{-1/2} A M^{-1/2}) \ll \kappa_2(A).$$

下半: 算特征值

2 (六) 特征值问题

基础

▷ 方法总览

{	极端特征对 $\rightarrow$ 幂法 / 反幂法,
	全部特征值 $\rightarrow$ QR 法,
	对称矩阵 $\rightarrow$ 三对角化后 QR / Jacobi,
	指定区间特征值 $\rightarrow$ 对称三对角上的 Sturm 二分法.

▷ Schur 分解

$$U^* A U = T, \quad t_{ii} = \lambda_i(A).$$

实矩阵对应实 Schur 形:

$$Q^\top A Q = \begin{bmatrix} \boxed{\times} & * & * \\ & \boxed{\times \times} & * \\ & & \ddots \end{bmatrix}.$$

幂法

▷ 幂法

$$A = V \Lambda V^{-1}, \quad |\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|,$$

$$x_0 = \alpha_1 v_1 + \dots + \alpha_n v_n, \quad \alpha_1 \neq 0.$$

则

$$A^k x_0 = \lambda_1^k \left[\alpha_1 v_1 + \sum_{j=2}^n \alpha_j \left(\frac{\lambda_j}{\lambda_1} \right)^k v_j \right].$$

归一化后: $\text{span}\{x_k\} \rightarrow \text{span}\{v_1\},$

$$\rho(x_k) = \frac{x_k^* A x_k}{x_k^* x_k} \rightarrow \lambda_1.$$

$$\text{收敛因子: } \left| \frac{\lambda_2}{\lambda_1} \right|.$$

$$\begin{cases} y_{k+1} = A x_k, \\ x_{k+1} = \frac{y_{k+1}}{\|y_{k+1}\|}. \end{cases}$$

▷ 反幂法与位移反幂法

反幂法 = 对 A^{-1} 作幂法

$$\begin{cases} A y_{k+1} = x_k, \\ x_{k+1} = \frac{y_{k+1}}{\|y_{k+1}\|}. \end{cases}$$

$$\lambda_i(A^{-1}) = \frac{1}{\lambda_i(A)} \Rightarrow \text{主导条件下}$$

收敛到模最小特征值对应方向.

不显式求 A^{-1} , 每步解线性方程组.

A 可逆.

位移反幂法为

$$\begin{cases} (A - \mu I) y_{k+1} = x_k, \\ x_{k+1} = \frac{y_{k+1}}{\|y_{k+1}\|}. \end{cases}$$

$$\lambda_i((A - \mu I)^{-1}) = \frac{1}{\lambda_i - \mu}.$$

$\Rightarrow$ 主导条件下

收敛到最接近 μ 的特征值对应方向.

$$\text{收敛因子: } \left| \frac{\lambda_* - \mu}{\lambda_{\text{next}} - \mu} \right|.$$

$$\mu \neq \lambda_i(A).$$

$$\mu_k = \frac{x_k^* A x_k}{x_k^* x_k},$$

Rayleigh 商迭代: μ_k 取上式, Hermitian 简单特征值附近局部常三次收敛.

QR 方法

▷ QR 迭代

$$A_k = Q_k R_k, \quad A_{k+1} = R_k Q_k.$$

$$A_{k+1} = Q_k^\top A_k Q_k,$$

$$\Rightarrow \sigma(A_{k+1}) = \sigma(A_k).$$

$$A_k = \hat{Q}_k^\top A_0 \hat{Q}_k, \quad \hat{Q}_k = Q_0 Q_1 \cdots Q_{k-1}.$$

极限形 = 实 Schur 形.

▷ Hessenberg 化

$$Q_0^\top A Q_0 = H, \quad H = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{bmatrix}.$$

若 $A = A^\top$, 则

$H = T$ 为对称三对角矩阵.

$$\text{预处理: } O(n^3), \quad H: O(n^2), \quad T: O(n).$$

▷ 带位移 QR

$$H_k - \mu_k I = Q_k R_k,$$

$$H_{k+1} = R_k Q_k + \mu_k I = Q_k^\top H_k Q_k.$$

$$\mu_k \approx \lambda \Rightarrow h_{n,n-1} \rightarrow 0 \text{ 更快.}$$

▷ 分裂

若某个次对角元满足

$$|h_{i+1,i}| \leq u(|h_{ii}| + |h_{i+1,i+1}|),$$

$$h_{i+1,i} = 0.$$

$$H = \begin{bmatrix} H_{11} & H_{12} \\ 0 & H_{22} \end{bmatrix}, \quad \sigma(H) = \sigma(H_{11}) \cup \sigma(H_{22}).$$

$$H = \begin{bmatrix} H_{n-1} & * \\ 0 & \lambda_n \end{bmatrix}, \Rightarrow \lambda_n \text{ 直接分离.}$$

▷ **Wilkinson 位移与 Francis 双位移**

$$C = \begin{bmatrix} a & b \\ b & d \end{bmatrix}, \quad \delta = \frac{a-d}{2},$$

Wilkinson 位移取更接近 d 的特征值, 即

$$\mu = d - \frac{\operatorname{sgn}(\delta)b^2}{|\delta| + \sqrt{\delta^2 + b^2}}.$$

$$\operatorname{sgn}(0) = 1.$$

一般实 Hessenberg 情形取

$$C = \begin{bmatrix} h_{n-1,n-1} & h_{n-1,n} \\ h_{n,n-1} & h_{nn} \end{bmatrix}.$$

$$s = \operatorname{tr} C, \quad t = \det C.$$

再用 Francis 双位移多项式

$$p(H) = H^2 - sH + tI.$$

$p(H)e_1$ 起步, Householder 追鼓包.

3 (七) 对称特征值问题**QR 方法**▷ **对称 QR 算法**

$$A = A^\top.$$

$$Q_0^\top A Q_0 = T,$$

$$T = \begin{bmatrix} \alpha_1 & \beta_1 & & \\ \beta_1 & \alpha_2 & \ddots & \\ & \ddots & \ddots & \beta_{n-1} \\ & & \beta_{n-1} & \alpha_n \end{bmatrix}.$$

$$T_k - \mu_k I = Q_k R_k,$$

$$T_{k+1} = R_k Q_k + \mu_k I = Q_k^\top T_k Q_k.$$

$$C = \begin{bmatrix} a & b \\ b & d \end{bmatrix}, \quad \delta = \frac{a-d}{2}.$$

$$\mu = d - \frac{\operatorname{sgn}(\delta)b^2}{|\delta| + \sqrt{\delta^2 + b^2}}.$$

$$\operatorname{sgn}(0) = 1.$$

$$|h_{i+1,i}| \leq \varepsilon(|h_{ii}| + |h_{i+1,i+1}|).$$

$$h_{i+1,i} = 0,$$

$$h_{i+1,i} = 0 \Rightarrow \text{分裂}.$$

Jacobi 方法▷ **Jacobi 方法**

$$A = A^\top.$$

$$J = \begin{bmatrix} I & & \\ & c & s \\ & & I \\ & -s & c \end{bmatrix}, \quad A^+ = J^\top A J,$$

$$\tan 2\theta = \frac{2a_{pq}}{a_{qq} - a_{pp}}, \quad c = \cos \theta, \quad s = \sin \theta.$$

$$\begin{cases} \tau = \frac{a_{qq} - a_{pp}}{2a_{pq}}, \\ t = \frac{\operatorname{sgn}(\tau)}{|\tau| + \sqrt{1 + \tau^2}}, \\ c = \frac{1}{\sqrt{1 + t^2}}, \\ s = tc. \end{cases}$$

$$\operatorname{sgn}(0) = 1.$$

$$\operatorname{off}(A)^2 = \sum_{i \neq j} a_{ij}^2,$$

$$\operatorname{off}(A^+)^2 = \operatorname{off}(A)^2 - 2a_{pq}^2.$$

$$A_k \rightarrow \Lambda, \quad V = J_1 J_2 \cdots, \quad A = V \Lambda V^\top.$$